

n0 QUIC NAT Traversal

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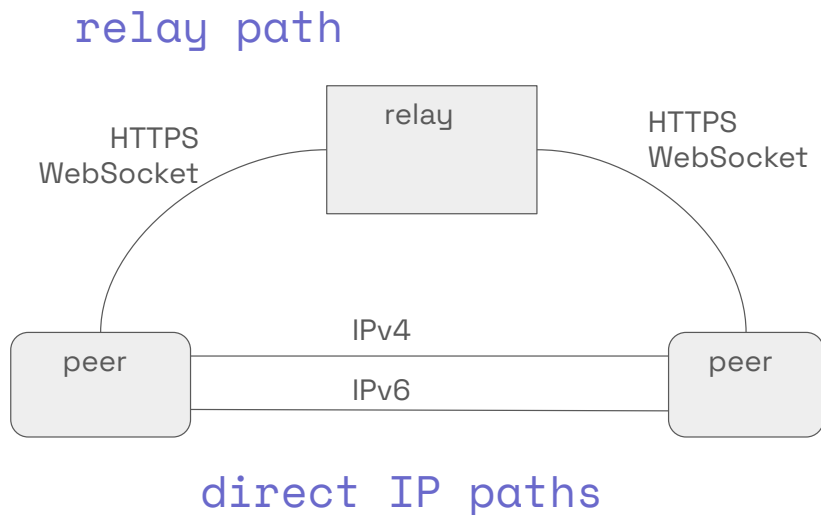
Diva Martinez

number 0

draft-bruynooghe-n0-quic-nat-traversal-00

<https://github.com/n0-computer/noq>

architecture



- Multipath negotiated
- NAT traversal coordinated over *path 0*
 - the relay path
- NAT probing using PATH_CHALLENGE
PATH_RESPONSE frames
- On successful probe
 - open *path 1...n*

n0 QUIC NAT Traversal

Based on draft-seemann-quic-nat-traversal

Server advertises address candidates:

```
ADD_ADDRESS Frame {  
    Type (i) = 0x3d7f90..0x3d7f91,  
    Sequence Number (i),  
    [ IPv4 (32) ],  
    [ IPv6 (128) ],  
    Port (16),  
}
```

Server revokes candidates

```
REMOVE_ADDRESS Frame {  
    Type (i) = 0x3d7f94,  
    Sequence Number (i),  
}
```

Clients initiates NAT traversal

```
REACH_OUT Frame {  
    Type (i) = 0x3d7f92..0x3d7f93,  
    round (i),  
    [ IPv4 (32) ],  
    [ IPv6 (128) ],  
    Port (16),  
}
```

- flood all address candidates
- no ICE address pairing
- off-path probing

Abandoned design

on-path NAT Probing on dedicated path IDs

- Server was still only using one path ID
 - (No address pairing)
- Too much path churn
- Consumes too many CIDs

Lack of path ID in NAT traversal

=> multipath not required

Challenges & Further work

- Issuing CIDs
 - Currently NAT probes use the **in-use active CID** 🦴
 - NAT probing consumes **a lot** of CIDs on one path
 - each path has to accept all CIDs
 - Interleaving PATH_NEW_CONNECTION_ID with REACH_OUT *may* work
 - If not: extension to negotiate a shared key?
 - generate CIDs at the peer
- Probing packets in multipath
- Connection establishment on multiple paths
 - In case a relay connection is not needed or impossible